

GUI

Class 1
Full-time studies

I. Without modifying the *main* method, make the code below work.

Hint: *Subject* class can have access to students' set (e.g. an array). Adding too many students to the set may throw an exception.

```
public class Main {
    public static void main(String args[]) {

        Person p1 = new Person("Jan", 50);
        Student s1 = new Student("Jasiek", 20);
        Person p3 = (Person)s1;

        p1.sayHelloTo(p3); //Hi Jasiek!
        p3.sayHelloTo(p1); //Hi Jan!

        Subject s = new Subject("GUI");
        s.setTeacher(p1);

        try {
            s.addStudent(s1);
        } catch (TooManyStudentsException e) {
            e.printStackTrace();
        }

        System.out.println(s); //GUI, teacher: Jan, students: Jasiek
    }
}
```

II. The code below depicts a class with a main method and what was printed by it on the console. The classes used in the main method are missing. Write them so that the code will work as intended. Make sure to proof your knowledge of inheritance in Java.

```
public class Task {
    public static void main(String[] args) {
        Laptop laptop = new Laptop("AMD", 64, "PPJ", 12345);
        Computer[] computers = new Computer[]{new Computer("i7", 24), laptop, new
            PersonalComputer("i5", 10.5, "PJA")};

        for (int i = 0; i < computers.length; i++) {
            System.out.println(computers[i]);
        }

        try {
            laptop.checkWeight();
        } catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Output:

```
ID: 2, type of chip: i7, memory size: 24.0GB
ID: 1, type of chip: AMD, memory size: 64.0GB, username: PPJ, weight: 12345.0g
ID: 3, type of chip: i5, memory size: 10.5GB, username: PJA
This laptop is not ergonomic - too heavy, over 5000g!
```